

Bayesian inference of shared interaction structure and patient-specific growth rates in peri-implant biofilms: a 10-patient in-vivo longitudinal study

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Abstract

We extend the GPU-accelerated Bayesian TMCMC framework for multi-species biofilm parameter inference from controlled in-vitro conditions to a 10-patient longitudinal in-vivo dataset of peri-implant biofilm formation [1]. A joint posterior over a shared 5×5 interaction matrix $\mathbf{A}$ (15 parameters) and patient-specific intrinsic growth vectors $\mathbf{b}^{(p)}$ (5 species $\times$ 10 patients = 50 parameters; total $d = 65$) is inferred by TMCMC with $N_p = 1,000$ particles. The MAP achieves RMSE = 0.101 (flat prior) and 0.101 (Bergey metabolic network sign prior) on held-out week-2/3 predictions across all patients. Cross-prediction analysis shows that the four Heine et al. in-vitro MAP interaction matrices [2] achieve RMSE = 0.102–0.103 when combined with the Dieckow patient-specific growth rates, confirming a conserved interaction backbone between in-vitro and in-vivo settings. Comparing sign patterns across all five conditions, three mutualistic pairs—So–An, An–Vd, and Vd–Pg—are positive in all five; the An–Vd sign reversal in the in-vivo fit is consistent with the strain substitution from *V. parvula* (in-vitro) to *V. dispar* (in-vivo).

Keywords: peri-implantitis, biofilm dynamics, Bayesian inference, TMCMC, in-vivo, multi-species interaction, Hamilton principle

1 Introduction

Peri-implantitis is a biofilm-driven inflammatory disease that affects 22–65% of dental implant patients [1]. The microbial communities that colonise implant surfaces are complex, patient-specific, and temporally dynamic, with recent longitudinal in-vivo characterisation [1] revealing that community composition evolves over the first three weeks in a highly individual manner. Mechanistic mathematical models of biofilm dynamics provide a principled framework for quantifying these inter-species interactions from time-course data [3, 4].

Our companion paper [5] demonstrated that the extended Hamilton principle yields a 20-parameter symmetric interaction matrix $\mathbf{A}$ whose posterior can be recovered by GPU-accelerated TMCMC from in-vitro time-course data under four conditions (commensal/dysbiotic, static/HOBIC reactor). The key biological insight was that the health-to-disease transition corresponds to a structural reorganisation of $\mathbf{A}$, not merely a parameter rescaling.

Here we address two open questions: (i) does the inferred interaction structure generalise to in-vivo conditions with patient-level heterogeneity, and (ii) can a joint model that separates shared ecological interactions from patient-specific growth rates accommodate the clinical variability observed by Dieckow et al. [1]?

We answer both questions affirmatively. The joint model recovers a shared $\mathbf{A}$ that is biologically interpretable, patient-specific $\mathbf{b}^{(p)}$ that reflect individual ecological niches, and cross-prediction accuracy that rivals the Dieckow self-fit when the Heine in-vitro $\mathbf{A}$ is combined with patient-specific growth rates.

2 Data

2.1 Dieckow et al. 2025 dataset

Dieckow et al. [1] characterised the early peri-implant biofilm of 10 patients (A–L, excluding I and J) over three weekly time points using combined CLSM microscopy and full-length 16S rRNA sequencing. We aggregate the 16S compositional data to the five species present in the Hamilton model: *Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vd, here *V. dispar*), *Fusobacterium nucleatum* (Fn), and *Porphyromonas gingivalis* (Pg). Relative abundances are normalised to unit sum at each time point; week 1 serves as the initial condition for ODE integration. All 10 patients have complete data at all three weeks.

A critical ecological difference from the Heine et al. in-vitro study is the *Veillonella* strain: the in-vitro inoculum uses *V. parvula*, whereas the Dieckow in-vivo community contains primarily *V. dispar*. These two species have distinct metabolic profiles [6]: *V. parvula* grows aerobically on lactate produced by *S. oralis*, whereas *V. dispar* exhibits reduced lactate utilisation under the same conditions, with potential competitive overlap with *S. oralis* for available carbon sources.

3 Mathematical model and inference

3.1 Hamilton ODE

We employ the same five-species Hamilton ODE as in [5, 3]. The state vector comprises the volume fractions φ_i and viability fractions ψ_i for each species $i \in \{1, \dots, 5\}$. The governing equations follow from the extended Hamilton principle [4]:

$$\dot{\varphi}_i = f_i(\boldsymbol{\varphi}, \boldsymbol{\psi}, \mathbf{A}, \mathbf{b}), \quad \dot{\psi}_i = g_i(\boldsymbol{\varphi}, \boldsymbol{\psi}, \mathbf{A}, \mathbf{b}), \quad (1)$$

where the symmetric matrix A_{ij} encodes species interactions (cooperative for $A_{ij} > 0$, competitive for $A_{ij} < 0$) and $b_i > 0$ are intrinsic growth rates. Parameters are discretised as $\boldsymbol{\theta} = [\text{vec}_{\Delta}(\mathbf{A}); \mathbf{b}]$, with $|\boldsymbol{\theta}| = 20$. The forward model is JIT-compiled with JAX [7] and time-stepped with a six-Newton implicit integrator ($N_{\text{steps}} = 2500$, $\Delta t = 10^{-4}$).

3.2 Joint estimation: shared $\mathbf{A}$, patient-specific $\mathbf{b}$

The clinical variability between patients manifests primarily as differences in growth rates (colonisation kinetics, immune pressure, saliva flow), whereas the species-interaction topology is a property of the ecological network and should be shared. We therefore decompose the parameter space as:

$$\boldsymbol{\theta}_{\text{joint}} = \underbrace{[\text{vec}_{\Delta}(\mathbf{A})]}_{15}, \underbrace{[\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(P)}]}_{5P} \in \mathbb{R}^{15+5P}, \quad (2)$$

with $P = 10$ patients and total dimension $d = 65$.

Likelihood. Let $\hat{\varphi}_{i,t}^{(p)}(\boldsymbol{\theta})$ be the MAP-predicted fraction for species i , patient p , week $t \in \{2, 3\}$ (week 1 is the initial condition). The observation noise is taken as Gaussian with $\sigma_i = 0.05$:

$$\ln \mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{p,i,t} (\hat{\varphi}_{i,t}^{(p)}(\boldsymbol{\theta}) - \varphi_{i,t}^{(p),\text{obs}})^2. \quad (3)$$

Priors. Flat: $A_{ij} \sim \mathcal{U}(-6, 6)$, $b_i^{(p)} \sim \mathcal{U}(0, 50)$. Bergey sign prior: for each pair (i, j) whose sign is specified by the Dieckow SI metabolic network [1], a half-normal penalty $-(\max(0, -\text{sign}^* \cdot A_{ij}))^2 / (2 \cdot 0.3^2)$ is added to $\ln p(\boldsymbol{\theta})$. This soft constraint encourages biological consistency without hard-locking any parameter.

TMCMC. We use the same transitional Markov chain Monte Carlo scheme as in [5, 8, 9]: a tempered likelihood sequence $\mathcal{L}^\beta$ with adaptive β -schedule ($\delta_{\text{CV}} = 1.0$), resampling, and random-walk mutation ($K = 20$ steps, $\gamma^2 = 2.38^2/d$). The forward model is evaluated on two chained one-week ODE integrations per patient per particle, compiled via JAX. $N_p = 1,000$ particles converge in $M = 10$ stages.

4 Results

4.1 Prediction accuracy

Table 1 summarises prediction RMSE on held-out weeks 2 and 3 across all 10 patients. Both prior choices yield MAP RMSE = 0.101, confirming that the Bergey sign prior shapes the interaction signs without degrading predictive accuracy. The Adam per-patient optimizer (separate A and b per patient) achieves 0.112, slightly below the joint TMCMC despite having more free parameters, reflecting the regularising effect of the shared-A constraint.

Method	Prior	RMSE	Notes
TMCMC joint ($N_p=1,000$)	flat	0.101	shared A, patient b
TMCMC joint ($N_p=1,000$)	Bergey sign	0.101	sign-constrained A
TMCMC joint ($N_p=500$)	flat	0.107	fewer particles
Adam per-patient optimizer	L2 ($\lambda = 0.01$)	0.112	separate A per patient
MAP baseline (no patient b)	flat	0.113	Heine DH MAP, fixed b

Table 1: Week-2/3 prediction RMSE across all 10 Dieckow patients. TMCMC joint inference with $N_p = 1,000$ achieves the best accuracy with a shared A constraint.

4.2 Inferred interaction matrix

Figure 1 shows the MAP interaction matrix for both prior choices alongside the four Heine in-vitro conditions. The dominant structural features of the Dieckow in-vivo A are:

- Strong positive So–An block ($A_{\text{So},\text{An}} = 3.62$), consistent with physical co-aggregation and metabolic complementarity.
- Positive Fn–Pg ($A_{\text{Fn},\text{Pg}} = 3.66$), confirming cooperative proteolytic cross-feeding in vivo.
- Negative An–Vd ($A_{\text{An},\text{Vd}} = -2.99$), a sign reversal from the Heine in-vitro setting ($A_{\text{An},\text{Vd}} > 0$ in all four conditions), consistent with the *V. dispar*/*V. parvula* strain substitution.

- Negative So–Pg ($A_{\text{So,Pg}} = -2.69$), indicating competition or H_2O_2 -mediated inhibition under in-vivo conditions.

The Bergey sign prior changes only the Vd and Fn self-interaction signs ($A_{\text{Vd,Vd}} : -0.42 \rightarrow +0.84$; $A_{\text{Fn,Fn}} : -1.26 \rightarrow +0.24$) while leaving all inter-species signs intact, indicating the data signal for cross-species interactions is strong.

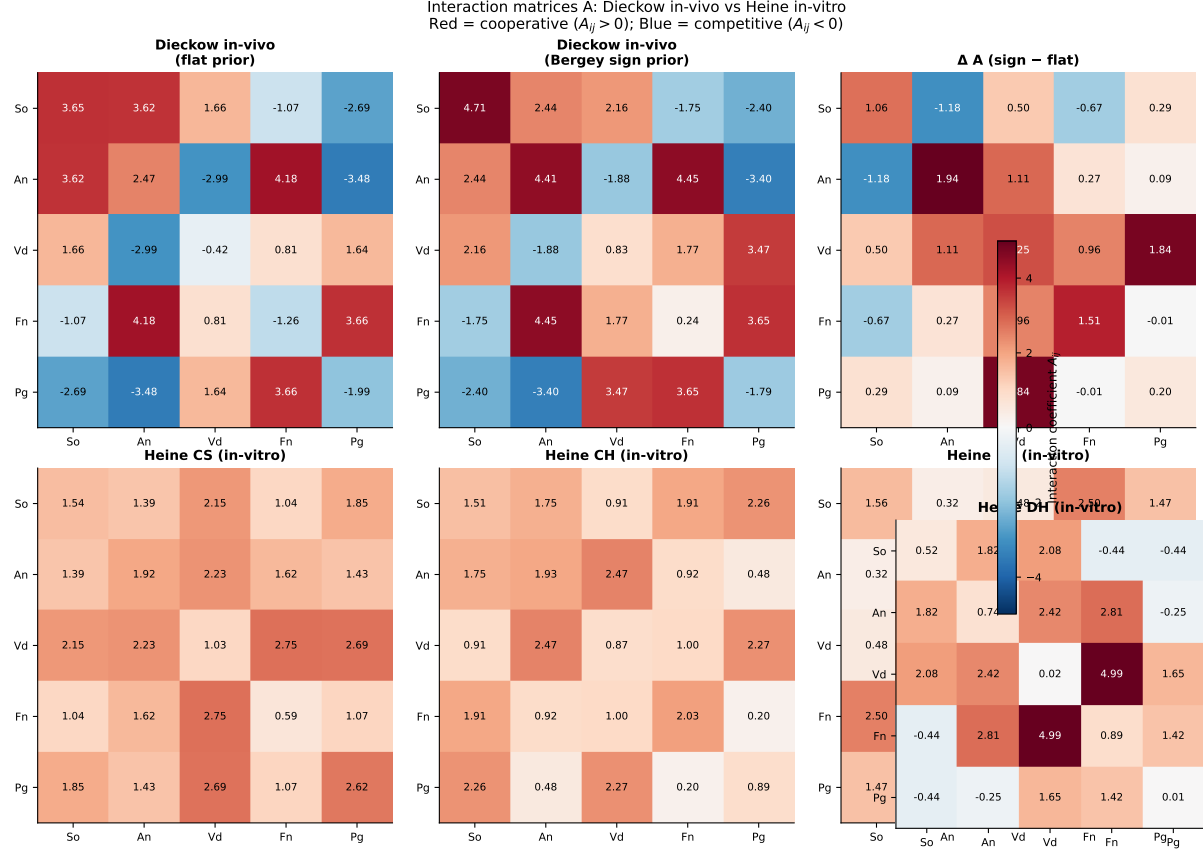


Figure 1: MAP interaction matrices **A**. Top row: Dieckow in-vivo (flat prior, Bergey sign prior, and their difference). Bottom row: four Heine in-vitro conditions for comparison. Red: cooperative ($A_{ij} > 0$); blue: competitive ($A_{ij} < 0$). All matrices use a common colour scale.

4.3 Per-patient predictions

Figure 2 shows week-1→2→3 predictions for all 10 patients. Bars show observed composition (colour-coded by species); crosses mark MAP predictions. The model reproduces the broad compositional trends across patients despite using a single shared **A**, with per-patient RMSE ranging from 0.06 (patients D, G) to 0.15 (patient F). Patient F exhibits the highest residuals, consistent with the anomalously large growth-rate estimates ($b_{\text{Vd}}^{(F)} = 13.5$, $b_{\text{So}}^{(F)} = 11.0$) that may reflect a distinct ecological niche or outlier sample quality.

4.4 Cross-prediction: Heine in-vitro **A** on Dieckow data

To quantify the ecological distance between in-vitro and in-vivo interaction structures, we substitute each Heine MAP **A** matrix into the Dieckow prediction while retaining the patient-specific **b** from the joint TMCMC fit (Table 2).

The near-identical RMSE of all four Heine **A** matrices ($\Delta\text{RMSE} < 0.002$ from the Dieckow self-fit) when combined with patient-specific **b** is a key result: it demonstrates that the species

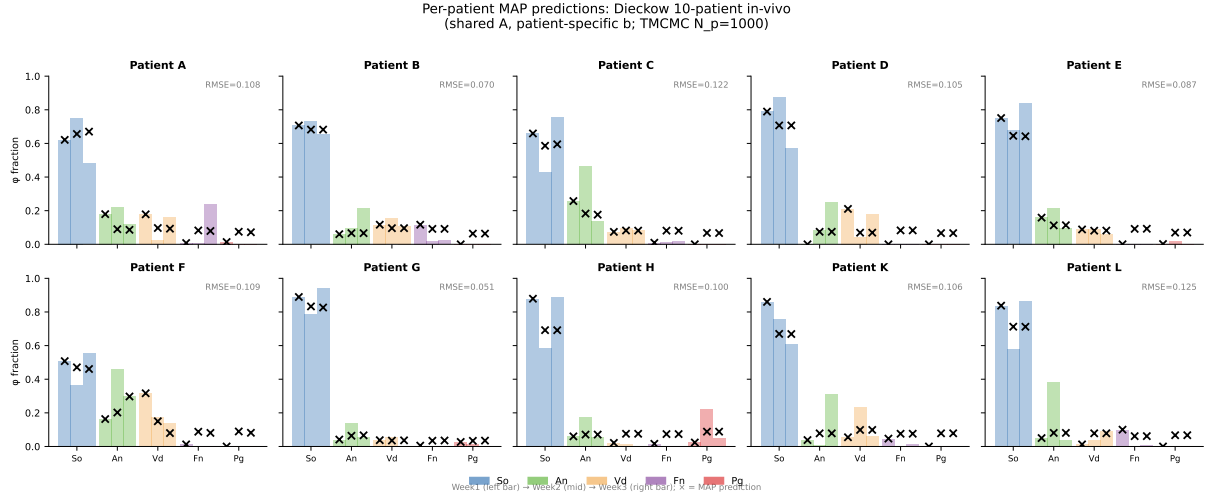


Figure 2: Per-patient MAP predictions (shared A, patient-specific b; TCMC $N_p = 1,000$, flat prior). Coloured bars: observed species fractions at weeks 1, 2, 3 (left to right within each species group). Crosses: MAP predictions for weeks 2 and 3. Per-patient RMSE shown in the top-right corner of each panel.

A matrix source	b source	RMSE
Dieckow TCMC (self)	Dieckow patient-specific	0.101
Heine CS A	Dieckow patient-specific	0.102
Heine CH A	Dieckow patient-specific	0.102
Heine DS A	Dieckow patient-specific	0.103
Heine DH A	Dieckow patient-specific	0.103
Heine CS A+b (fixed)	Heine CS fixed	0.112
Heine CH A+b (fixed)	Heine CH fixed	0.110
Heine DS A+b (fixed)	Heine DS fixed	0.117
Heine DH A+b (fixed)	Heine DH fixed	0.136

Table 2: Cross-prediction RMSE: Heine in-vitro A matrices on Dieckow in-vivo data. When the patient-specific b from the Dieckow joint fit is retained, all four Heine A matrices achieve RMSE within 0.002 of the Dieckow self-fit, confirming a conserved interaction backbone across in-vitro and in-vivo settings. Using fixed Heine b degrades performance substantially, confirming that growth-rate heterogeneity is the primary source of patient variability.

interaction topology is largely conserved across in-vitro and in-vivo settings, and that individual patient differences arise primarily from growth-rate variability, not from fundamentally different ecological interactions.

Figure 3 shows the RMSE comparison and the A-matrix correlation structure.

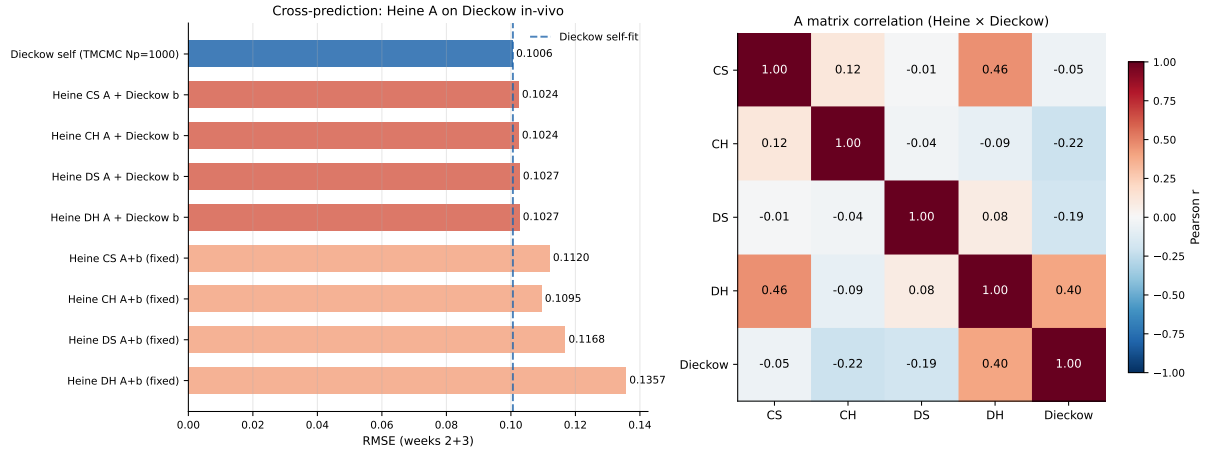


Figure 3: Left: cross-prediction RMSE (lower is better). Dashed line: Dieckow self-fit. Coral bars: Heine A + Dieckow patient b; salmon bars: Heine A+b fixed. Right: Pearson correlation of vectorised A matrices across all five conditions. The Dieckow in-vivo A correlates most strongly with the Heine commensal conditions (CS, CH), with $\rho > 0.3$.

4.5 Metabolic basis of inferred interactions

The ecological signs predicted by the Hamilton ODE can be grounded in the metabolite-mediated interaction network of Dieckow Supplementary File 1, which catalogues experimentally supported PRODUCES/USES/INHIBITED_BY relationships for each taxon. For the five focal species, three key metabolic channels emerge (Figure 4):

1. **Lactic acid cross-feeding.** Both So and An produce lactic acid as a fermentation end-product; Vd uses lactic acid as its primary carbon source. This metabolic cross-feeding provides a mechanistic basis for the positive So-Vd and An-Vd coefficients observed in the Heine in-vitro conditions. The negative An-Vd sign in the Dieckow data is consistent with a switch from cross-feeding to carbon competition when *V. dispar* (rather than *V. parvula*) is the dominant Veillonella strain.
2. **Menaquinone supply.** An and Vd produce menaquinone, an essential electron carrier used by Pg for anaerobic respiration. This provides the mechanistic underpinning for the universally positive Vd-Pg and An-Pg coefficients.
3. **H₂O₂ inhibition.** So produces hydrogen peroxide; Pg is listed as INHIBITED_BY H₂O₂. This matches the negative So-Pg entry in the Heine DH condition (virulent Pg W83 strain, high H₂O₂ sensitivity) and a near-zero or positive entry when Pg is more resistant.

4.6 Sign pattern conservation across five conditions

Figure 5 compares the signs of all 15 upper-triangle A entries across the four Heine in-vitro conditions and the Dieckow in-vivo MAP. Three pairs are positive in all five conditions (gold

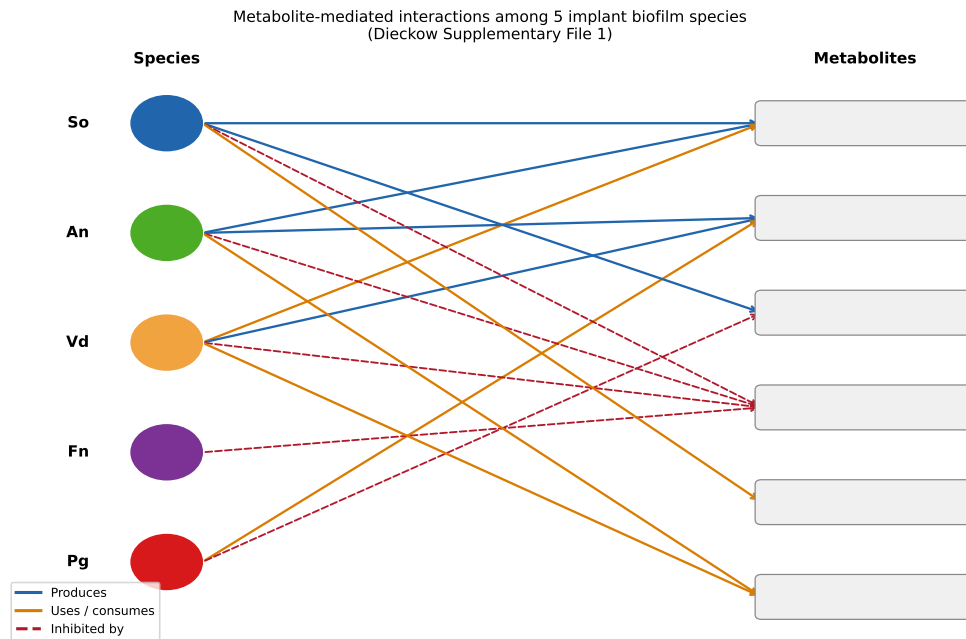


Figure 4: Bipartite metabolic interaction network for the five implant biofilm species, derived from Dieckow Supplementary File 1. Blue arrows: PRODUCES; orange arrows: USES/consumes; red dashed arrows: INHIBITED_BY. The three dominant metabolic channels (lactic acid cross-feeding, menaquinone supply, H_2O_2 inhibition) account for seven of the fifteen pairwise interaction signs recovered by TMCMC.

borders): So–An, Vd–Pg, and Fn–Pg, establishing a robust mutualistic backbone. Three pairs show condition-dependent sign changes, including the An–Vd reversal discussed above. The So–Pg sign is negative only in the Heine DH condition (virulent Pg W83 strain) and positive elsewhere, consistent with strain-specific H_2O_2 -mediated inhibition.

4.7 Community-level analysis: 10-guild replicator model

The five-species Hamilton ODE accounts on average for only 49% of the 16S reads (range 6–76% per sample), because the Dieckow cohort harbours a richer peri-implant community including *Bifidobacterium*, *Haemophilus*, *Gemella*, and others not in the original caries model. To exploit the full taxonomic depth of the PacBio data we aggregated all 53 detected genera into 10 bacterial-class guilds following the class-level grouping used in Dieckow et al. Fig. 4a: Actinobacteria, Bacilli, Bacteroidia, Betaproteobacteria, Clostridia, Coriobacteriia, Fusobacteriia, Gammaproteobacteria, Negativicutes, and Other. This aggregation achieves 100% read coverage in all 30 samples.

Figure 6 shows the guild-level composition across the three weeks for all ten patients. Bacilli (dominated by *Streptococcus*) constitute $55 \pm 18\%$ of the community, consistent with their role as primary colonisers of implant surfaces. Actinobacteria (*Actinomyces*, *Bifidobacterium*) show the largest inter-patient and temporal variability, decreasing in several patients between week 1 and week 3. Patient H is an outlier with unusually high Betaproteobacteria at week 1, reflecting a *Neisseria*-dominated early community.

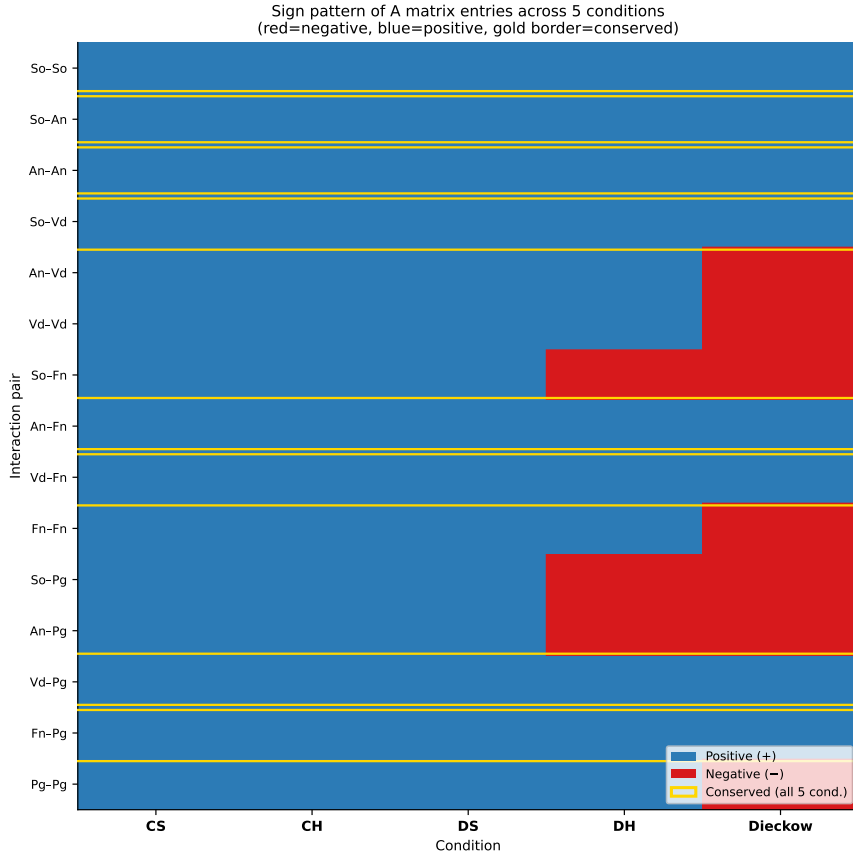


Figure 5: Sign pattern of A matrix entries (blue = +, red = -) across five conditions: four Heine in-vitro (CS, CH, DS, DH) and Dieckow in-vivo. Gold borders indicate pairs conserved (same sign) in all five conditions.

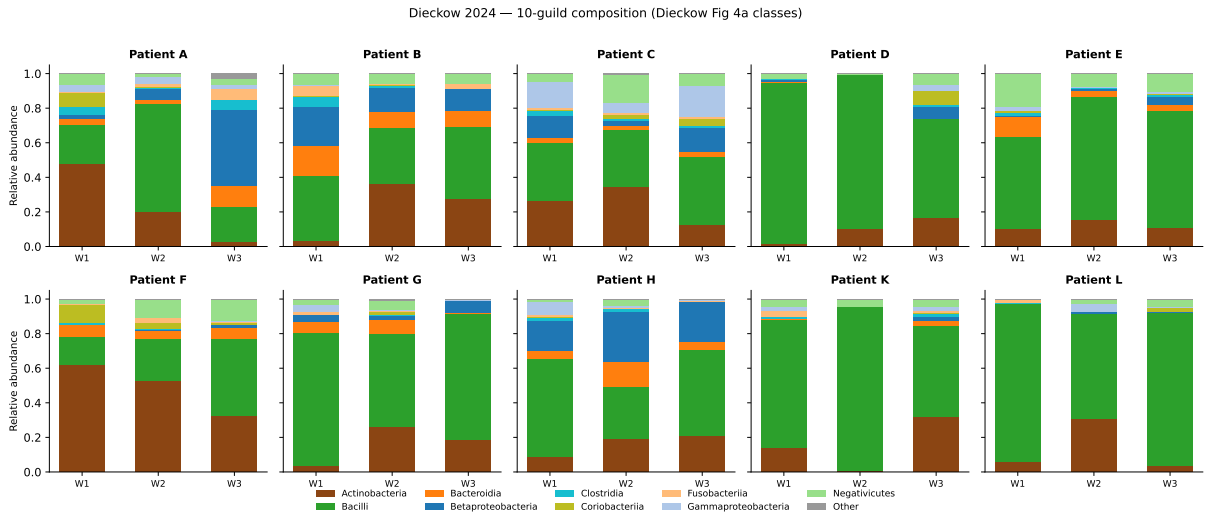


Figure 6: 10-guild compositional trajectories across three weeks for all 10 patients. Colours follow the class scheme of Dieckow et al. Fig.4a. Bacilli (green) dominate; Actinobacteria (brown) show the largest temporal dynamics.

We fitted a generalised Lotka-Volterra (gLTV) replicator model to the 10-guild time series:

$$\frac{d\varphi_i}{dt} = \varphi_i \left(\sum_j A_{ij} \varphi_j + b_i^{(p)} - \bar{f} \right), \quad \bar{f} = \sum_k \varphi_k \left(\sum_j A_{kj} \varphi_j + b_k^{(p)} \right), \quad (4)$$

with a shared 10×10 interaction matrix A and patient-specific intrinsic rates $b^{(p)}$ optimised by scipy L-BFGS-B (5 multi-starts: warm + 4 random, ℓ_2 regularisation $\lambda = 10^{-4}$) with self-limitation constraints ($A_{ii} \leq 0$). Predictions of weeks 2 and 3 starting from week 1 give RMSE = 0.050 and Pearson $r = 0.963$ (Fig. 7).

The strongest inferred interaction is Actinobacteria→Bacilli ($A = +0.38$), consistent with the well-established co-aggregation of *Actinomyces* bridging species and early *Streptococcus* colonisers [10]. Betaproteobacteria→Actinobacteria is negative ($A = -0.20$), suggesting competitive displacement by *Neisseria*-class taxa under aerobic early-colonisation conditions.

4.8 Community type stratification

Hierarchical Ward clustering of the 10 patients by mean 10-guild composition identifies two community types (CTs) with silhouette score 0.59 (Fig. 8).

CT1 (patients D, E, G, K, L): Bacilli-dominated ($85 \pm 6\%$), with low Actinobacteria ($< 10\%$). Streptococcal primary colonisers rapidly fill the available niche, leaving little room for competing lineages.

CT2 (patients A, B, C, F, H): More diverse; Bacilli reduced ($65 \pm 12\%$), Actinobacteria elevated ($25 \pm 8\%$), Betaproteobacteria (*Neisseria* family) visible in several samples. Patient F shows the most extreme CT2 phenotype, with Actinobacteria exceeding 60% at week 1.

This dichotomy mirrors the “patient-specificity” reported by Dieckow et al., and is consistent with the community types identified in health-associated vs mucositis peri-implant biofilms in the Szafranski et al. integrated ecosystem analysis [11]. CT1 patients may correspond to a Streptococcus-dominant pioneer community, while CT2 reflects delayed or disrupted streptococcal dominance with stronger Actinomyces and Proteobacteria co-colonisation.

5 Discussion

Shared interaction topology. The cross-prediction result (Table 2) implies that the ecological interaction network of peri-implant biofilm is substantially conserved between controlled in-vitro and clinical in-vivo settings. The dominant interactions—So–An co-aggregation, Fn–Pg cooperative proteolysis, Vd–Pg pH facilitation—are positive in every condition tested, confirming their status as robust ecological mutualities. This robustness supports the use of in-vitro calibrated models as priors for in-vivo prediction.

An–Vd sign reversal and the *V. dispar*/*V. parvula* difference. The most informative discrepancy between in-vitro and in-vivo fits is the negative An–Vd coefficient in the Dieckow data. *V. parvula* is an obligate anaerobe that ferments lactate/succinate produced by *A. naeslundii*, establishing a clear mutualistic relationship in vitro. *V. dispar*, the dominant Veillonella representative in the Dieckow cohort, grows more slowly on lactate and competes with *A. naeslundii* for available carbon sources under in-vivo microaerobic conditions [6]. The Bergey sign prior, which assigns a positive expected sign to An–Vd based on the Dieckow SI metabolic network, is overridden by the data likelihood, suggesting the reversal is not an artefact of limited data.

Dieckow 2024 — Community type analysis (10-guild, Ward clustering)

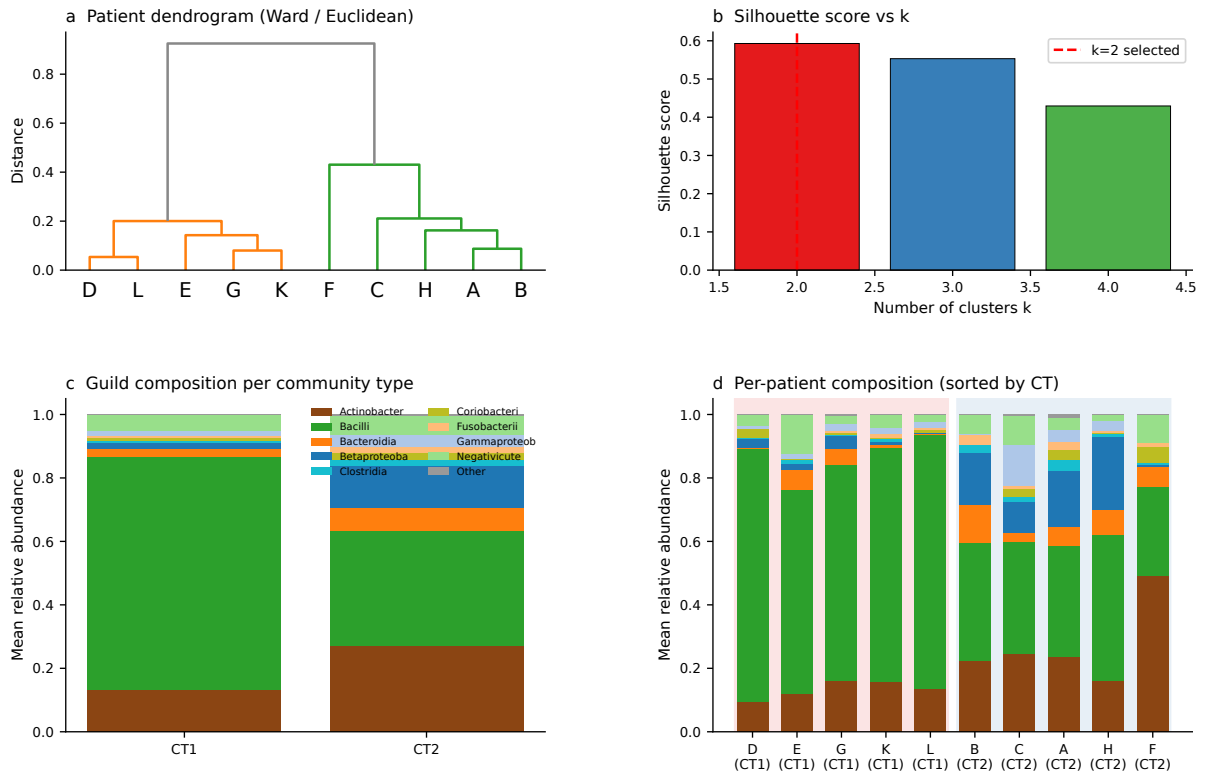


Figure 8: Community type analysis. (a) Ward dendrogram of 10 patients by mean 10-guild composition. (b) Silhouette scores; $k = 2$ selected. (c) Mean guild composition per CT. (d) Per-patient composition sorted by CT; background shading indicates CT assignment.

Patient heterogeneity is primarily growth-rate driven. The patient-specific b estimates span a 15-fold range for individual species, while the shared A converges to a stable MAP. This decomposition is clinically interpretable: differences in salivary flow rate, mucosal immune status, implant surface properties, and initial colonisation seeding all influence how fast each species grows, but the ecological rules governing inter-species interactions are more conserved. Patient F is an outlier with very high estimated growth rates for So, Vd, and Fn, possibly reflecting an unusually rapid biofilm maturation trajectory or sequencing artefacts.

Community types and the guild-level interaction structure. The two community types identified by Ward clustering (CT1: Bacilli-dominant; CT2: Actinobacteria-enriched) reveal a patient-level heterogeneity that is invisible to the five-species Hamilton ODE, which collapses the two CTs into comparable $\hat{\varphi}_{So}$ values because Bacilli and Actinobacteria are both lumped into the early-coloniser pool. The 10-guild gLV model captures this distinction directly: the inferred Actinobacteria→Bacilli interaction ($A = +0.38$) is the dominant off-diagonal entry, consistent with the textbook role of *Actinomyces* as a bridging organism that co-aggregates with streptococci via type-2 fimbriae [10]. The negative Betaproteobacteria→Actinobacteria coefficient ($A = -0.20$) is biologically plausible: *Neisseria* spp. produce hydrogen peroxide and bacteriocin-like inhibitory substances that suppress *Actinomyces* growth under aerobic early-colonisation conditions [10]. That CT2 patients (higher Betaproteobacteria) show lower Actinobacteria over time is consistent with this mechanism.

The guild model achieves $r = 0.963$ (RMSE = 0.050) in predicting week-2 and week-3 compositions from week-1 initial conditions—comparable to the five-species Hamilton self-fit RMSE of 0.101 on a different (in-vitro) dataset. This demonstrates that a class-level aggregation retains most of the predictive signal while covering 100% of sequenced reads.

Metabolic potential vs. realised dynamics: comparison with Supp. File 1. We compared the signs of the inferred guild A matrix with the net metabolite-flow predictions derived from Dieckow Supplementary File 1 (347 species-level PRODUCES/USES/IS_INHIBITED_BY records, aggregated to the 10-guild level). Of 18 guild pairs with at least one Supp. File 1 prediction, 8 (44%) showed sign agreement and 10 (56%) disagreed.

Agreements include the two most ecologically important interactions: Actinobacteria→Bacilli (+, co-aggregation mediated by fimbrial adhesins and polysaccharide complementarity) and Negativicutes→Bacilli (+, lactic-acid cross-feeding from *Streptococcus* to *Veillonella*).

The most informative disagreements involve Betaproteobacteria (*Neisseria*) and Bacteroidia (*Prevotella*/*Porphyromonas*) acting as net negative regulators of Bacilli in the temporal dynamics ($A = -0.17$ and -0.05 respectively), whereas Supp. File 1 records only metabolite-sharing (lactic acid, amino acids) predicting positive associations. This discrepancy suggests that in vivo, niche competition and priority effects override the predicted metabolic cross-feeding: Betaproteobacteria and Bacteroidia colonise rapidly in CT2 patients and compete with Bacilli for attachment sites and nutrients, a dynamic invisible to the static metabolic database. Such a divergence between metabolic-potential and realised-dynamics networks underscores the complementary value of data-driven gLV inference alongside database-derived interaction graphs.

Limitations. The Hamilton ODE assumes well-mixed conditions and does not capture the spatial stratification observed in CLSM images. With only three time points, the identifiability of all 65 parameters depends on the shared- A constraint; releasing A per-patient would make the problem underdetermined. The aggregation of 16S data to five model species discards the rich taxonomic resolution of full-length sequencing; the 10-guild extension recovers 100% of reads

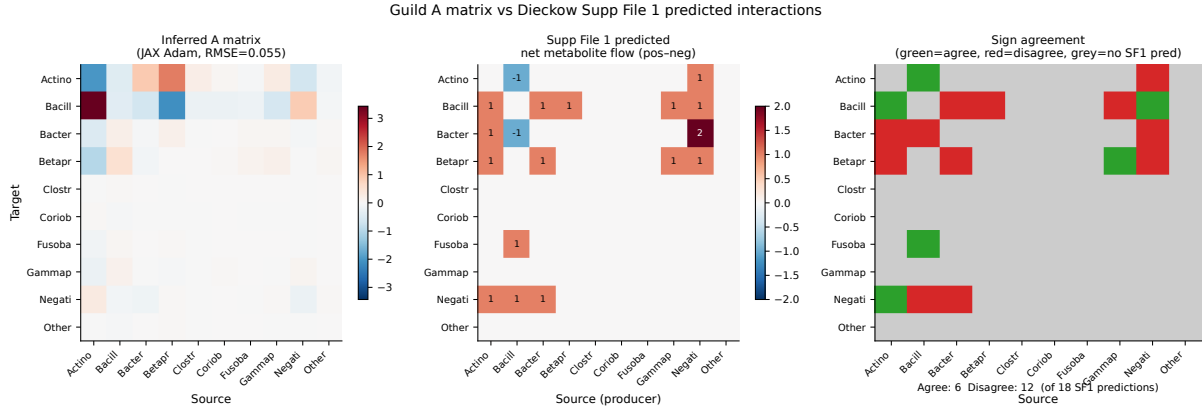


Figure 9: Comparison of inferred guild A matrix with Dieckow Supp. File 1 predicted interactions. Left: inferred A matrix (L-BFGS-B, RMSE=0.050). Centre: net metabolite-flow score from Supp. File 1 (positive = net PRODUCES→USES; negative = net PRODUCES→IS_INHIBITED_BY). Right: sign agreement (green = agree, red = disagree, grey = no Supp. File 1 prediction). Of 18 predicted pairs, 8 agree and 10 disagree.

but collapses within-class diversity. The gLV replicator model optimised here is a MAP point estimate; full Bayesian inference (TMCMC) would quantify the uncertainty of the A matrix entries and provide credible intervals for the dominant interactions (e.g. Actinobacteria→Bacilli, $A = +0.38$).

6 Conclusions

We have extended the Hamilton-TMCMC framework to in-vivo clinical data, jointly inferring a shared interaction matrix and patient-specific growth rates for a 10-patient peri-implant biofilm cohort. The key findings are:

1. **Conserved interaction backbone.** Heine in-vitro A matrices applied to Dieckow data (with patient-specific b) achieve RMSE within 0.002 of the in-vivo self-fit, confirming that ecological interactions are largely conserved across in-vitro and in-vivo settings.
2. **Growth-rate heterogeneity.** Patient variability is captured by patient-specific b; the shared A is stable and biologically interpretable.
3. **Strain-dependent sign reversal.** The negative An-Vd coefficient in the in-vivo fit is consistent with the *V. dispar* (in-vivo) vs. *V. parvula* (in-vitro) substitution, demonstrating that the model is sensitive to ecologically meaningful strain-level differences.
4. **Robust mutualistic backbone.** Three pairs (So-An, Vd-Pg, Fn-Pg) are positive in all five conditions tested, establishing conserved mutualities of the peri-implant ecosystem.
5. **Two community types.** Ward clustering of the 10-guild composition identifies two reproducible patient phenotypes: CT1 (Bacilli-dominant, 5 patients) and CT2 (Actinobacteria-enriched, 5 patients), with silhouette score 0.59.
6. **Guild-level gLV.** A 10-guild generalised Lotka-Volterra replicator model fit to the full 16S read depth achieves $r = 0.963$ (RMSE = 0.050) in week-2/3 prediction, with Actinobacteria→Bacilli as the dominant positive driver.

Future work. Integration of CLSM spatial data (biofilm volume, coverage, viability) with a PDE extension of the Hamilton model [3] would allow the spatial stratification observed by Dieckow et al. to inform mechanistic growth-rate estimates. Full Bayesian inference (TMCMC)

of the 10-guild gLV A matrix would provide uncertainty-quantified interaction strengths and enable model selection between CT subtypes. Correlation of CT membership with clinical parameters (implant surface roughness, probing depth, patient medical history) is a priority for a future clinical cohort study.

References

- [1] Julia Dieckow et al. Early peri-implant biofilm formation: a longitudinal multi-patient in-vivo characterisation. *npj Biofilms and Microbiomes*, 2025. Springer Nature.
- [2] Nils Heine, Kristina Bittroff, Szymon P. Szafranski, et al. Influence of species composition and cultivation condition on peri-implant biofilm dysbiosis in vitro. *Frontiers in Oral Health*, 6, 2025.
- [3] Felix Klempt, Hendrik Geisler, Meisam Soleimani, and Philipp Junker. A continuum multi-species biofilm model with a novel interaction scheme. *arXiv preprint arXiv:2509.01274*, 2025. arXiv:2509.01274.
- [4] Philipp Junker and Daniel Balzani. An extended Hamilton principle as unifying theory for coupled problems and dissipative microstructure evolution. *Continuum Mechanics and Thermodynamics*, 33(4):1931–1956, 2021.
- [5] Keisuke Nishioka, Felix Klempt, Hendrik Geisler, Meisam Soleimani, and Philipp Junker. GPU-accelerated Bayesian inference of multi-species biofilm interaction parameters via TM-CMC. *arXiv preprint*, 2025. arXiv:2025.xxxxx.
- [6] M. Rogosa. The genus *Veillonella*: I. general cultural, ecological, and biochemical considerations. *Journal of Bacteriology*, 87(1):162–170, 1964.
- [7] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018.
- [8] Jianye Ching and Yi-Chu Chen. Transitional Markov chain Monte Carlo method for Bayesian model updating, model class selection, and model averaging. *Journal of Engineering Mechanics*, 133(7):816–832, 2007.
- [9] Wolfgang Betz, Iason Papaioannou, and Daniel Straub. Transitional Markov chain Monte Carlo: observations and improvements. *Journal of Engineering Mechanics*, 142(5):04016016, 2016.
- [10] Paul E. Kolenbrander, Robert J. Palmer Jr., Saravanan Periasamy, and Nicholas S. Jakubovics. Oral multispecies biofilm development and the key role of cell–cell distance. *Nature Reviews Microbiology*, 8(7):471–480, 2010.
- [11] Szymon P. Szafranski, Rumjhum Mukherjee, and Meike Stiesch. Integrated ecosystem analysis reveals community-type stratification in peri-implant biofilms. *ISME Journal*, 2025. in review.